

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [3]: df = pd.read_csv("Social_Network_Ads.csv")
df
```

```
Out[3]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [4]: df.drop(columns=["User ID"], inplace=True)
df["Gender"] = df["Gender"].map({"Male": 1, "Female": 0})
df
```

Out[4]:

	Gender	Age	EstimatedSalary	Purchased
0	1	19	19000	0
1	1	35	20000	0
2	0	26	43000	0
3	0	27	57000	0
4	1	19	76000	0
...	...	...	...	...
395	0	46	41000	1
396	1	51	23000	1
397	0	50	20000	1
398	1	36	33000	0
399	0	49	36000	1

400 rows × 4 columns

```
In [5]: x = df[["Gender", "Age", "EstimatedSalary"]]
        y = df["Purchased"]
```

```
In [6]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.20, random_s
```

```
In [7]: scaler = StandardScaler()
        x_train = scaler.fit_transform(x_train)
        x_test = scaler.transform(x_test)
```

```
In [8]: model = LogisticRegression()
        model.fit(x_train, y_train)
```

```
Out[8]: LogisticRegression ⓘ ?
        ► Parameters
```

```
In [9]: y_pred = model.predict(x_test)
```

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In [10]: accuracy = accuracy_score(y_test, y_pred)
         accuracy
```

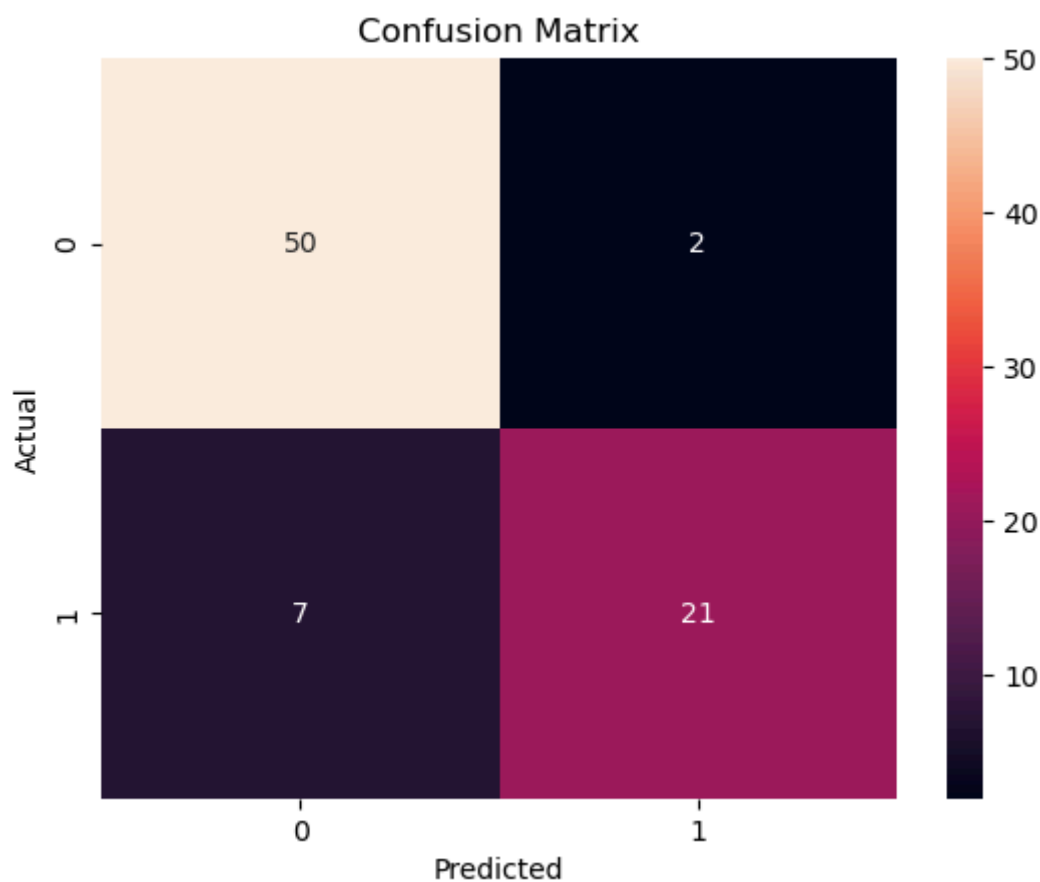
Out[10]: 0.8875

```
In [12]: conf_matrix = confusion_matrix(y_test, y_pred)
         conf_matrix
```

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Out[12]: array([[50,  2],
               [ 7, 21]])
```

```
In [15]: sns.heatmap(conf_matrix, annot = True,)
         plt.xlabel("Predicted")
         plt.ylabel("Actual")
```

```
plt.title("Confusion Matrix")  
plt.show()
```



```
In [16]: new_data = np.array([[1, 30, 80000]])  
new_data_df = pd.DataFrame(new_data, columns = x.columns)  
new_data_scaled = scaler.transform(new_data_df)  
prediction = model.predict(new_data_scaled)  
if prediction == 1:  
    print("Purchase Prediction: Yes")  
else:  
    print("Purchase Prediction: No")
```

Purchase Prediction: No

In []: